

Koorosh Azizi

Assistant Professor

Tennessee State University



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[Google Scholar](#)



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My research examines how infrastructure, human behavior, and institutions interact to shape the reliability and equity of urban water and flood-risk systems.

APPOINTMENTS

Jan. 2026–Present	Assistant Professor Tennessee State University - Department of Civil and Architectural Engineering
Jun. 2024–Jan. 2026	Postdoctoral Fellow The University of Texas at Austin - Jackson School of Geosciences and LBJ School of Public Affairs
Jun. 2022–Jun. 2024	Postdoctoral Research Scholar Arizona State University - School of Sustainability
Sep. 2017–May 2022	Graduate Research Assistant The University of Memphis - Department of Civil Engineering and Department of City and Regional Planning, Memphis, TN

EDUCATION

Sep. 2017–May 2022	Ph.D., Civil and Environmental Engineering The University of Memphis, Memphis, TN, USA
Sep. 2014–Jan. 2017	M.S., Civil and Environmental Engineering Shahid Beheshti University, Tehran, Iran
Sep. 2010–Sep. 2014	B.S., Civil Engineering Isfahan University of Technology, Isfahan, Iran

PEER-REVIEWED PUBLICATIONS

1. **Azizi, K.**, Garcia, M., Niyogi, D., & Bixler, R. P. (2026). Coupled human–environmental systems: Socio-behavioral and spatial dynamics in green stormwater infrastructure adoption. *Nature Cities*. In press.
2. **Azizi, K.**, Bixler, R. P., & Passalacqua, P. (2026). Gaps and opportunities for collaborative flood governance: Network-analytic insights from the US Gulf Coast. *Water Resources Research*, 62, e2025WR041974.
3. **Azizi, K.**, & Bixler, R. P. (2026). Flood exposure and household green stormwater infrastructure adoption. *Environmental Research Communications*, 8(2), 025016.
4. **Azizi, K.**, Wang, Y., Enriquez, O., Passalacqua, P., Niyogi, D., & Bixler, R. P. (2025). Integrating perspectives: Multi-sectoral insights into U.S. Gulf Coast flood governance. *International Journal of Disaster Risk Reduction*, 127, 105662.
5. **Azizi, K.**, Barnes, J. L., Deslatte, A., Koebele, E. A., Anderies, J. M., & Garcia, M. (2025). Balancing effectiveness and equity in sustainable water management transitions: Case of the Miami-Dade Water and Sewer Department. *Journal of Water Resources Planning and Management*, 151(1), 04024058.
6. **Azizi, K.**, Barnes, J. L., Anderies, J. M., & Garcia, M. (2024). Equity implications of efficient water conservation programs. *Environmental Research Letters*, 19(9), 094015.
7. **Azizi, K.**, Hornberger, G. M., Baggio, J., Koebele, E. A., Anderies, J. M., & Garcia, M. (2024). Identifying conditions that support the provision of high-quality and affordable urban drinking water in the US. *Journal of Water Resources Planning and Management*, 150(8), 04024024.

8. Wiechman, A., Alonso Vicario, S., Anderies, J. M., Garcia, M., **Azizi, K.**, & Hornberger, G. M. (2024). Institutional dynamics impact the response of urban socio-hydrologic systems to supply challenges. *Water Resources Research*, 60(2), e2023WR035565.
9. Ebrahimi, S., **Azizi, K.**, Khorram, M., Kashani, A. R., & Ali, A. S. A. (2024). Evaluation of hydrological variabilities of water quality models considering event-based scenarios: A case study. *Stochastic Environmental Research and Risk Assessment*, 38(6), 2425–2449.
10. **Azizi, K.**, Diko, S. K., & Meier, C. I. (2023). A citizen science approach to the characterisation and modelling of urban pluvial flooding. *Water Alternatives*, 16(1), 265–294.
11. **Azizi, K.**, Diko, S. K., Saija, L., Zamani, M. G., & Meier, C. I. (2022). Integrated community-based approaches to urban pluvial flooding research, trends and future directions: A review. *Urban Climate*, 44, 101237.
12. **Azizi, K.**, & Meier, C. I. (2022). Urban pluvial flood risk assessment: Challenges and opportunities for improvement using a community-based approach. In *World Environmental and Water Resources Congress 2021: Planning a Resilient Future along America's Freshwaters* (pp. 350–361). American Society of Civil Engineers.
13. **Azizi, K.**, Kashani, A. R., Ebrahimi, S., & Jazaei, F. (2022). Application of a multi-objective optimization model for the design of piano key weirs with a fixed dam height. *Canadian Journal of Civil Engineering*, 49(11), 1764–1778.
14. Kashani, A. R., Camp, C. V., **Azizi, K.**, & Rostamian, M. (2022). Multi-objective optimization of mechanically stabilized earth retaining wall using evolutionary algorithms. *International Journal for Numerical and Analytical Methods in Geomechanics*, 46(8), 1433–1465.
15. Kashani, A. R., Gandomi, A. H., **Azizi, K.**, & Camp, C. V. (2022). Multi-objective optimization of reinforced concrete cantilever retaining wall: A comparative study. *Structural and Multidisciplinary Optimization*, 65(9), 262.
16. Kashani, A. R., Camp, C. V., Rostamian, M., **Azizi, K.**, & Gandomi, A. H. (2022). Population-based optimization in structural engineering: A review. *Artificial Intelligence Review*, 55(1), 345–452.
17. **Azizi, K.**, Attari, J., & Moridi, A. (2017). Estimation of discharge coefficient and optimization of piano key weirs. In *Labyrinth and Piano Key Weirs III: PKW 2017* (pp. 213–220). CRC Press.

MANUSCRIPTS UNDER REVIEW

1. Garcia, M., Deslatte, A., Koebele, E. A., Hornberger, G. M., Anderies, J. M., Alonso Vicario, S., **Azizi, K.**, Barnes, J., & Wiechman, A. (2026). Design for dynamic fitness: Archetypes of urban water systems. Manuscript under review at *Environmental Research: Infrastructure and Sustainability*. Preprint available at arXiv:2601.17147.
2. Siegel, H. G., Bahamón-Pinzón, D., **Azizi, K.**, Banner, J. L., Almanza, S., & Atkinson, L. (2026). Participatory research: A panacea for science skepticism in the geosciences? Manuscript under review at *Nature Geoscience*.
3. Wang, Y., **Azizi, K.**, & Bixler, R. P. (2026). Communities and conditions: What supports green stormwater infrastructure adoption? Manuscript under review at the *Journal of Environmental Management*.
4. Ebrahimi, S., Khorram, M., **Azizi, K.**, & Kumar, S. (2026). Earth observation and data assimilation in arid, irrigation-dominated, and transboundary basins: A review. Manuscript under review at the *Journal of Hydrology*.
5. Alonso Vicario, S., Wiechman, A., **Azizi, K.**, Anderies, J. M., & Garcia, M. (2026). Vulnerability of urban water systems to changing droughts: The case of Santa Rosa, California. Manuscript under review at *Environmental Research: Infrastructure and Sustainability*.

MANUSCRIPT IN PREPARATION

1. **Azizi, K.**, Dashtian, H., Garcia, M., Niyogi, D., Bixler, R. P., Bates, P. D., & Scanlon, B. (2026). Different Hazards, Different NFIP Protection Gaps Across the United States. Manuscript in preparation.

CONFERENCE PRESENTATIONS (FIRST AUTHOR ONLY)

1. **Azizi, K.**, Dashtian, H., Garcia, M., Niyogi, D., Bixler, R. P., Bates, P. D., & Scanlon, B. R. (2026). Diagnosing the hazard- and location-specific challenges of the National Flood Insurance Program. Abstract submitted to the *AGU26 Annual Meeting*.
2. **Azizi, K.**, Garcia, M., Niyogi, D., & Bixler, R. P. (2025). Socio-behavioral dynamics of household stormwater management: An agent-based exploration of green stormwater infrastructure adoption. Presented at the *AGU25 Annual Meeting*, New Orleans, Louisiana. Abstract SY41A-07.
3. **Azizi, K.**, Passalacqua, P., & Bixler, R. P. (2024). Mechanisms of flood risk governance networks: Insights from Southeast Texas. Presented at the *AGU24 Annual Meeting*. Abstract SY51B-2580.
4. **Azizi, K.**, Barnes, J. L., Deslatte, A., Koebele, E. A., Anderies, J. M., & Garcia, M. (2023). Equity and effectiveness in sustainable water management practices: Insights from Miami-Dade's water conservation program. Presented at the *AGU23 Fall Meeting*. Abstract GC43I-1409.
5. **Azizi, K.**, Hornberger, G. M., Baggio, J., Koebele, E. A., Anderies, J. M., & Garcia, M. (2023). Identifying key factors for providing high-quality and affordable drinking water: A study of U.S. urban water systems. Presented at the *AGU23 Fall Meeting*. Abstract H33E-01.
6. **Azizi, K.**, & Meier, C. I. (2022). Improving urban pluvial flooding characterization and modeling through a citizen science approach. Presented at the *AGU22 Fall Meeting*. Abstract H15T-1032.
7. **Azizi, K.**, & Meier, C. I. (2022). Citizens' contributions improve modeling of urban pluvial flooding: A case study. Presented at the *World Environmental and Water Resources Congress 2022*, Atlanta, Georgia.
8. **Azizi, K.**, Meier, C. I., & Saija, L. (2020). Improving the characterization of urban flash floods through the application of local knowledge. Presented at the *AGU Fall Meeting 2020*. Abstract H162-0011.
9. **Azizi, K.**, Meier, C. I., & Saija, L. (2019). Applying a socio-hydrological approach to understanding and managing urban stormwater: A case study in North Memphis. Presented at the *AGU Fall Meeting 2019*. Abstract H11O-1732.
10. **Azizi, K.**, Meier, C. I., & Saija, L. (2019). Stormwater management at the local level: Participatory community-based strategies and mitigation practices. Presented at the *World Environmental and Water Resources Congress 2019*, Pittsburgh, Pennsylvania.
11. **Azizi, K.**, Meier, C. I., & Saija, L. (2018). Bottom-up planning: From natural disaster to community disaster. Presented at the *AGU Fall Meeting 2018*. Abstract PA23F-1032.
12. **Azizi, K.**, Attari, J., & Moridi, A. (2017). Estimation of discharge coefficient and optimization of piano key weirs. Presented at the *Third International Workshop on Labyrinth and Piano Key Weirs*, Qui Nhon, Vietnam.

GRANTS, FELLOWSHIPS, AND SPONSORED RESEARCH

PENDING PROPOSALS

- **2026 | Principal Investigator**
Urban Storm Response Across Scales: Where Satellite Signals Become Hydrologic Information. NASA ROSES-2025 F.17 Research Initiation Awards. **\$257,512**. Submitted and under review.
- **2026 | Principal Investigator**
Quantifying the Impacts of Natural Disasters on Tennessee's Transportation Assets: A Consequence-Pathway Method to Rank, Explain, and Update Resilience Priorities. Tennessee Department of Transportation, FY2026 Problem Statement #2. **\$273,285**. Submitted and under review. Co-Principal Investigator: Deo Chimba.
- **2026 | Co-Investigator**
LACCE Geospatial and Mechanical-Hydrological Landslide Models Characterizing the Regional Influence of Extreme Precipitation on Slope Stability in California. NASA ROSES-25 A.6, Landslide Change Characterization Experiment Science Team. **\$630,435**. Submitted and under review. Principal Investigator: Amber Spears.

Koorosh Azizi - Curriculum Vitae

PROPOSALS IN PREPARATION

- **2026 | Principal Investigator**
Excellence in Research: From Installations to Delivered Protection: Coupled Human-Infrastructure Network Dynamics in Distributed Green Stormwater Infrastructure. National Science Foundation, Historically Black Colleges and Universities Excellence in Research (HBCU-EiR). **Approximately \$900,000**. Full proposal in preparation. Proposed subaward collaborator: Dev Niyogi, The University of Texas at Austin.
- **2026 | Principal Investigator**
Resilience Between Actions: Interaction Signatures and Collective Adaptation Pathways in Urban Flood Systems. National Science Foundation, CMMI Infrastructure Systems and People. **Approximately \$1.1 million collaborative request**. Proposal in preparation. Co-Principal Investigator: R. Patrick Bixler, The University of Texas at Austin.

AWARDED GRANTS AND FELLOWSHIPS

- **2026 | Principal Investigator**
Transportation Network Vulnerability to Roadway Flooding in Nashville: A Coupled Hazard-Network-Behavior Framework for Emergency Access. CHDT/University Transportation Center Research Grant. **\$54,000**. Awarded; project period Aug. 2026–Jul. 2027.
- **2026 | Faculty Awardee**
UJIMA Faculty Award. Tennessee State University. **\$2,500**. Awarded.
- **2018 | Co-Principal Investigator**
Urban Flood Resilience in Memphis. Internal Seed Grant, The University of Memphis. **\$50,000**. Awarded.
- **2017–2020 | Graduate Fellow**
Herff Graduate Fellowship. Herff College of Engineering, The University of Memphis. **\$105,000**. Competitive three-year doctoral fellowship including a monthly stipend and health insurance.

SELECTED RECENT PROPOSALS ACTIVITY

- **2026 | Principal Investigator**
Timing and Targeting Green Stormwater Infrastructure Incentives: A Coupled Socio-Hydrologic Framework for Runoff Reduction and Water-Quality Improvement. U.S. Geological Survey 104b Program and Tennessee Department of Environment and Conservation. **\$84,000**. Submitted; not funded.
- **2026 | Principal Investigator**
Coupled Energy-Water Infrastructure Dynamics Under Compound Heat and Flood Extremes: A Mechanism-Based Modeling and Evaluation Framework. U.S. Department of Energy, Biological and Environmental Research, Earth and Environmental Systems Modeling. **\$875,000**. Pre-application submitted; not selected.
- **2025 | Co-Principal Investigator**
Deploying AI for Smarter and Sustainable Irrigation Systems for Public. Good Systems Seed Funding Program, The University of Texas at Austin. **\$50,000**. Not funded.

ADDITIONAL AWARDS AND TRAVEL SUPPORT

- American Geophysical Union Conference Travel Grant, 2019, \$1,200
- World Environmental and Water Resources Congress Travel Grant, 2019, \$1,000
- American Geophysical Union Conference Travel Grant, 2018, \$1,200
- World Environmental and Water Resources Congress Travel Grant, 2018, \$1,000

ACADEMIC TEACHING EXPERIENCE

Tennessee State University, Department of Civil and Architectural Engineering **Instructor of Record**

Fall 2026

- Hydrology
- Introduction to Environmental Engineering
- Fluid Mechanics
- Fluid Mechanics Laboratory

Spring 2026

- Hydraulic Engineering
Student evaluation: **4.55/5.00**
- Water and Wastewater Engineering
Student evaluation: **4.98/5.00**

The University of Texas at Austin, LBJ School of Public Affairs

- **Co-Instructor**, *Sustainable Cities: Human-Environment Systems*, Summer 2024

Arizona State University, School of Sustainable Engineering and the Built Environment

- **Guest Lecturer**, *Hydrology*, Fall 2022 and Fall 2023
- **Guest Lecturer**, *Sociohydrology*, Spring 2023

The University of Memphis, Department of Civil Engineering

- **Workshop Instructor**, *Urban Hydrological Modeling*, 2021–2022
- **Teaching Assistant**, *Hydrology*, 2019–2021
- **Laboratory Instructor**, *Hydrology Laboratory*, 2018–2020

Shahid Beheshti University, Department of Civil Engineering

- **Instructor**, *Hydraulic Design of Structures*, 2016–2017
- **Instructor**, *Hydraulics*, 2014–2016

INVITED TALKS AND LECTURES

1. Tennessee State University, Department Seminar Series, November 14, 2026. (*Scheduled*)
2. James Madison University, School of Integrated Sciences, Seminar Series, March 4, 2025.
3. Colorado State University, Department of Systems Engineering, Department Seminar Series, December 3, 2024.
4. University of South Florida, Department of Geosciences, Department Seminar Series, February 28, 2024.
5. Pennsylvania State University, Department of Geosciences, Department Seminar Series, November 27, 2023.
6. Arizona State University, School of Sustainable Engineering and the Built Environment, Hydrosocial Seminar Series, September 15, 2023.
7. Arizona State University, School of Sustainability, Sustainability Seminar Series, April 15, 2023.

SELECTED STUDENT RESEARCH MENTORSHIP

- **Emery Bell (B.S.)**, The University of Texas at Austin, Jackson School of Geosciences, *GIS Analysis of Risk Impacts*, Spring 2025–Spring 2026.
- **Sara Steele (B.S.)**, The University of Texas at Austin, Jackson School of Geosciences, *Understanding Precipitation Patterns*, Fall 2024–Spring 2025.

Koorosh Azizi - Curriculum Vitae

- **Yuer Wang (Ph.D.)**, The University of Texas at Austin, LBJ School of Public Affairs, *Qualitative and Quantitative Methods in Environmental Science*, Fall 2024–Present.
- **Sara Alonso Vicario (Ph.D.)**, Arizona State University, Civil Engineering, *Hydrological Analysis*, Fall 2022–Spring 2024.
- **Francesca Federico (Ph.D.)**, Arizona State University, Sustainability, *Risk Sharing and Climate Insurance Programs*, Spring 2023–Spring 2024.
- **Adam Wiechman (Ph.D.)**, Arizona State University, Sustainability, *Urban Water Systems and Dynamical Systems*, Fall 2022–Spring 2024.
- **Karissa Gund (M.S.)**, Arizona State University, Civil Engineering, *Hydrological Variability and Flow Regimes*, Fall 2022–Fall 2023.
- **Nischal Kafle (Ph.D.)**, The University of Memphis, Civil Engineering, *Urban Flood Management and Green Infrastructure*, Fall 2021–Fall 2023.
- **Samin Khorram (Ph.D.)**, The University of Memphis, Civil Engineering, *Watershed Modeling*, Fall 2021–Fall 2023.

SELECTED RESEARCH PROJECTS

- **Transportation Network Vulnerability to Roadway Flooding in Nashville** | Principal Investigator, Tennessee State University | 2026–2027
Examining how flood hazards, transportation infrastructure, traveler behavior, and emergency-response institutions interact to shape reliable and equitable access to critical services.
- **Coupled Human-Infrastructure Dynamics of Green Stormwater Infrastructure** | Lead Investigator, Tennessee State University and The University of Texas at Austin | 2023–Present
Examining how household behavior, social processes, drainage connectivity, and maintenance institutions jointly shape the reliability and equity of distributed stormwater protection.
- **Flood Insurance Protection Gaps Across the United States** | Lead Investigator | 2025–Present
Analyzing how physical flood risk, insurance institutions, and public protection interact to produce geographically distinct protection gaps and unequal household burdens.
- **Collaborative Flood Governance in the U.S. Gulf Coast** | Lead Investigator, The University of Texas at Austin | 2024–2026
Using stakeholder interviews, surveys, and network analysis to examine how institutional relationships and coordination shape the reliability and equity of regional flood-risk governance.
- **Project CRESSLE: Community Resilience Integrated into an Earth System Science Learning Ecosystem** | Researcher, The University of Texas at Austin | 2024–Present
Integrating atmospheric, land-cover, geospatial, and social-vulnerability data to examine how environmental hazards and community conditions shape unequal exposure and resilience to extreme heat and precipitation.
- **Transition Dynamics in Integrated Urban Drinking Water Systems** | Researcher, Arizona State University | 2022–2024
Combining system dynamics modeling, comparative utility analysis, and practitioner engagement to examine how infrastructure, institutions, and conservation behavior shape water-system reliability, affordability, and adaptation.
- **Community-Based and Citizen-Science Urban Flood Research** | Lead Researcher, The University of Memphis | 2018–2022
Integrating community knowledge, citizen science, field observations, and 1D–2D flood modeling to improve the reliability, legitimacy, and equity of urban flood assessment and resilience planning.

PEER-REVIEW SERVICE

- Scientific Reports
- Environmental Research Letters
- Environmental Research: Climate
- Environmental Research: Infrastructure and Sustainability
- Environmental Research: Water
- Environmental Research Communications
- Journal of Hydrology
- Journal of Water Resources Planning and Management
- Journal of Hydrologic Engineering
- Journal of Infrastructure Systems
- Theoretical and Applied Climatology
- Journal of Flood Risk Management
- Urban Climate
- Urban Water Journal
- Sustainable Cities and Society
- Discover Water
- Sustainability
- Urban Science
- Journal of Hydroinformatics

PROFESSIONAL MEMBERSHIPS AND AFFILIATIONS

- American Geophysical Union
- American Society of Civil Engineers

TECHNICAL SKILLS

Research and Analytical Methods: Statistical analysis; system dynamics modeling; agent-based modeling; network analysis; geospatial analysis; optimization; qualitative interviews; focus groups; and field research.

Hydrologic and Hydraulic Modeling: SWMM, PCSWMM, HEC-RAS, HEC-HMS, IBER, and FLO-2D.

Geospatial and Engineering Software: ArcGIS, QGIS, and AutoCAD.

Programming Languages: R, Python, MATLAB, and Julia.

LANGUAGES

- **English:** Fluent (spoken and written)
- **Persian:** Fluent (spoken and written)
- **Kurdish:** Native